

Part C / MSc

Bayes Methods

Mock 1 2018

Do not turn this page until you are told that you may do so

1. Suppose a group of N players are selected play K games in a club tennis tournament. The players have played each other before and are fairly evenly matched, but strongest players in the contest are very unlikely to loose against the weakest players.

In the Bradley-Terry model for the outcome of a tennis game, for $i, j \in \{1, \dots, N\}$, the probability $p_{i,j}$ that player i beats player j is

$$p_{ij} = \frac{\exp(\lambda_i)}{\exp(\lambda_i) + \exp(\lambda_j)}$$

where $\lambda_i \in \Re$ is a parameter measuring player i 's skill. Let $\lambda = (\lambda_1, \dots, \lambda_N)$.

For $k = 1, \dots, K$, let (i_k, j_k) denote the ordered pair of players in the k th game. Let $y_k \in \{0, 1\}$ give the outcome, win for i_k (1) or loss (0).

(a)

- (i) Elicit a prior for λ , noting any further validation you would undertake. *Most of the marks are for reasoning, few for the prior itself.*
- (ii) Write down the posterior $\pi(\lambda|y)$ in terms of the model elements.
- (iii) Give an MCMC algorithm targeting $\pi(\lambda|y)$.
- (iv) Briefly outline how you would carry out convergence analysis for your MCMC.

- (b) Suppose for player $i = 1, 2, \dots, N$ covariate information $x_i = (x_{i,1}, \dots, x_{i,p})$ is available, with $x_{i,1} = 1$ corresponding to the intercept and p a small number (say $p \simeq 4$). The covariates are all thought likely to be informative. Let $\beta = (\beta_1, \dots, \beta_p)^T$ be a parameter vector, and let $z = (z_1, \dots, z_p)^T$ be a model indicator vector. For $j = 1, 2, \dots, p$ let $z_j \in \{0, 1\}$. Let $\beta_z = (\beta_1 z_1, \dots, \beta_p z_p)^T$. Consider the Bradley-Terry model with covariates, in which $\lambda_i = x_i \beta_z$,

$$p_{ij} = \frac{\exp(x_i \beta_z)}{\exp(x_i \beta_z) + \exp(x_j \beta_z)}$$

- (i) Elicit a prior for z, β , noting any further validation you would undertake.
- (ii) Write down the posterior $\pi(z, \beta|y)$ in terms of the model elements.
- (iii) Give an MCMC algorithm targeting $\pi(z, \beta|y)$.
- (iv) How would you use the output of your algorithm in Q1biii to (A) test for an effect due to the j th covariate $(x_{1,j}, \dots, x_{N,j})$, (B) identify the unknown true set of explanatory covariates and (C) estimate the probability for player i to beat player j in a future game, if the appropriate loss function is the MSE?

2. (a) Suppose $Y \sim f(y|\theta)$ and $\theta \sim \pi(\theta)$ with θ a real scalar parameter. Suppose we observe $Y = y$ and wish to estimate θ .
- (i) Explain what it means for $\pi(\theta|y)$ to be *doubly intractable* and the consequence for MCMC-based estimation of θ given $Y = y$.
 - (ii) Let $S(y) = (S_1(y), \dots, S_p(y))$ be a p -component vector of statistics computed from the data. For $S, S' \in \mathbb{R}^p$ let $d(S, S')$ be a distance function and let $\rho > 0$ be a simulation parameter. Write down an MCMC-ABC algorithm simulating

$$\theta \sim \pi(\theta | d(S(Y), S(y)) < \rho).$$

- (b) Let $\phi = (\phi_1, \dots, \phi_m)^T$, $\phi \in [0, \infty)^m$ and $\psi = (\psi_1, \dots, \psi_n)^T$, $\psi \in [0, \infty)^n$ be column vectors of positive parameters. Let $\mu = \phi\psi^T$ denote the $m \times n$ outer product matrix. Consider the observation model

$$Y_{i,j} \sim \text{Poisson}(\mu_{i,j}), \quad i = 1, \dots, m, \quad j = 1, \dots, n,$$

and suppose data $Y = y$ are observed.

Thinking of the cells in the matrix μ as a lattice, denote by $N_{i,j}$ the set of labels (a, b) , $a \in 1 : m$, $b \in 1 : n$ for cells adjoining cell (i, j) (interior cells have 4 neighbors, and boundary cells 3, except at the corners where each cell has 2 neighbors). Consider the joint prior density

$$\pi(\phi, \psi | \theta) \propto \exp(-\alpha \bar{\phi} - \beta \bar{\psi}) \exp(-\theta \sum_{i=1}^m \sum_{j=1}^n \sum_{(a,b) \in N_{i,j}} |\phi_a \psi_b - \phi_i \psi_j|),$$

where $\bar{\phi} = m^{-1} \sum_{i=1}^m \phi_i$, $\bar{\psi} = n^{-1} \sum_{j=1}^n \psi_j$, α and β are fixed positive prior hyperparameters and $\theta \geq 0$ is unknown with prior $\pi(\theta)$.

- (i) Briefly explain why the posterior $\pi(\theta, \phi, \psi | y)$ is doubly intractable and write down the marginal posterior $\pi(\theta | y)$ in terms of the model elements.
- (ii) Consider the summary statistics $S = (\bar{y}_{1,:}, \dots, \bar{y}_{m,:}, \bar{y}_{:,1}, \dots, \bar{y}_{:,n})$ with

$$\begin{aligned} \bar{y}_{i,:} &= n^{-1} \sum_{j=1}^n y_{i,j}, \quad i = 1, \dots, m \\ \bar{y}_{:,j} &= m^{-1} \sum_{i=1}^m y_{i,j}, \quad j = 1, \dots, n. \end{aligned}$$

Briefly motivate this choice of the ABC statistics.

- (iii) Suppose an MCMC algorithm simulating $(\phi, \psi) \sim \pi(\phi, \psi | \theta)$ is available. Give an MCMC-ABC algorithm approximately simulating

$$\theta \sim \pi(\theta | d(S(Y), S(y)) < \rho).$$

3. (a) Let $x_1, x_2, x_3, \dots$ be an infinite sequence of binary random variables.
- (i) What does it mean to say that these variables are exchangeable?
 - (ii) What form does de Finetti's theorem give for the joint distribution $p(x_1, x_2, \dots, x_n)$?
 - (iii) Let $n_0 = \sum_{i=1}^n x_i$. Show that

$$p(x_1, \dots, x_n | n_0) = \binom{n}{n_0}^{-1}.$$

- (b) Consider a Markov chain $X_0, X_1, X_2, \dots$ with $X_n \in \{0, 1\}$, initial distribution $X_0 \sim \text{Bernoulli}(0.5)$ and transition matrix $P = \begin{pmatrix} 1-a & a \\ a & 1-a \end{pmatrix}$ with $0 \leq a \leq 1$. Let $\#x = \sum_{i=1}^n |x_i - x_{i-1}|$.
- (i) Show that $p_n(x_0, \dots, x_n) = \exp(\log(a/(1-a))\#x + n \log(1-a) - \log(2))$
 - (ii) For what values of a are the random variables $X_0, X_1, X_2, \dots$ exchangeable?
- (c) For $n = 1, 2, \dots$ let $z = (z_1, \dots, z_n)$ be jointly normal,

$$p_n(z_1, \dots, z_n) = N(z; 0_n, \Sigma^{(n)})$$

with 0_n a vector of n zeros, and $\Sigma_{i,i}^{(n)} = 1$ and $\Sigma_{i,j}^{(n)} = \rho$ for $i, j \in \{1, 2, \dots, n\}$, $i \neq j$.

- (i) Show that

$$p_n(z_1, \dots, z_n) = \int_{-\infty}^{\infty} \prod_{i=1}^n N(z_i; \theta, 1-\rho) N(\theta; 0, \rho) d\theta$$

- (ii) Prove that the distribution $p_n(z_1, \dots, z_n)$ exists for every n if and only if $0 \leq \rho < 1$.
Hint: the vector $(1, 1, \dots, 1)^T$ is an eigenvector of $\Sigma^{(n)}$.
- (iii) Show that the densities $p_n(z_1, \dots, z_n)$ are marginally consistent, that is,

$$p_n(z_1, \dots, z_n) = \int_{-\infty}^{\infty} p_{n+1}(z_1, \dots, z_n, z_{n+1}) dz_{n+1}.$$

- (iv) For each $i = 1, 2, 3, \dots$, we set $q_i = \Phi(z_i)$ and $x_i \sim \text{Bernoulli}(q_i)$ with Φ the cdf of a standard normal. Prove that the infinite binary sequence $x_1, x_2, x_3, \dots$ is an infinite exchangeable sequence.

(a) (i) $\exists i, j$ such that p_{ij} can be as small as " a " (a = "very unlikely" say $a = 1/1000$ = needs to be given).

Say m pairs are this extreme.

$$m = \sum_{i,j} \mathbb{I} p_{ij} < a$$

taking expectation over λ

$$m = \sum_{i,j} \Pr(p_{ij} < a)$$

Now if the pairs are a priori exchangeable

$\Pr(p_{ij} < a)$ same for all pairs

(marginally) $\Rightarrow$

$$\Pr(p_{ij} < a) \geq \frac{m}{\binom{N}{2}} \quad (*)$$

Take $\pi(\lambda) = \prod_{i=1}^N N(\lambda_i; 0, \sigma^2)$

[exchangeable, most players similar score, few extreme, could also argue skills λ_i is sum many factors $\rightarrow$ CLT]

Now use something $\lambda_i, \lambda_j \sim N(0, \sigma^2)$ and adjust σ^2 so that $(*)$ holds.

[Remark: in fact a simple calc

$$\text{suggests } \Pr(\lambda_i - \lambda_j < \log(\frac{a}{1-a})) = \frac{m}{\binom{N}{2}}$$

$$\text{so } \sigma = \frac{1}{\sqrt{2}} \log\left(\frac{a}{1-a}\right) \left[\Phi^{-1}(m/\binom{N}{2}) \right]^{-1}$$

$$(ii) \pi(\lambda | y) \propto \pi(\lambda) L(y; \lambda)$$

$$\pi(\lambda) = \prod_{i=1}^N N(\lambda_i; 0, \sigma^2)$$

$$L(\lambda; y) = \prod_{k=1}^K (p_{ik,jk})^{y_k} (1 - p_{ik,jk})^{1-y_k}$$

$$p_{ik,jk} = \frac{e^{\lambda_{ik}}}{e^{\lambda_{0k}} + e^{\lambda_{ik}}}$$

(iii) let $\Lambda_t = \lambda$. let ω be fixed $\omega > 0$.

For $i = 1 \dots N$

- set $\lambda' \leftarrow \lambda$, set $\lambda'_i \sim U(\lambda_i - \omega, \lambda_i + \omega)$

$$\text{- w.p. } \alpha(\lambda' | \lambda) = \frac{\pi(\lambda') L(\lambda'; y)}{\pi(\lambda) L(\lambda; y)}$$

set $\Lambda_{t+1} = \lambda'$ else $\Lambda_{t+1} = \lambda$

1 (iv) $\downarrow$

① plot traces of $\log |k|$ & $|X^{(k)}; y|$ against t . Look for evidence of trend.

② Make multiple runs (3?) check marginal dbrs are stable across runs.

③ Calculate ESS check variance estimates at desired precision.

④ Plot ACF check falls off to zero at asymptotic variation.



1 (b) Reasonable chance any given covariate is informative ("not thought likely").

$P(z_i = 1) = \frac{1}{2}$ will do this. Not given info on covariates (eg are some similar?) so assume any combination of 4 equally likely.

$$\pi_1(\beta) \propto N(\beta; 0, \sigma_\beta^2) \quad \pi_2(z) \propto 1$$

Now use simulation to check p_{ij}

has similar dbr to that in (a), adjust σ_β to achieve this.

1 (6) (ii)

$$f_1(z, \beta | y) \propto L(\beta; z; y) \pi(\beta | z) \pi_2(z)$$

$$\propto L(\beta; z; y) \pi(\beta) \propto 1$$

$L(\beta; z; y)$ as before with

$$p_{ik} = \frac{e^{x_{ik} \beta z}}{e^{x_{ik} \beta z} + e^{x_{jk} \beta z}}$$

(iii) let $z_2 = z$, $\beta_1 = \beta$, $w > 0$ be given.

for i in $1 \dots p$

① - set $z' \leftarrow z$, set $z'_i \leftarrow 1 - z_i$

$$\text{- w.p. } \alpha(\beta, z' | \beta, z) \propto 1 \wedge \frac{L(\beta; z'; y)}{L(\beta; z; y)}$$

set $z_{t+1} = z'$ else set $z_{t+1} = z$.

② ~~set~~ $\beta' \leftarrow \beta$, $\beta'_i \sim U(\beta_i - w, \beta_i + w)$

$$\text{- w.p. } \alpha(\beta', z | \beta, z) \propto 1 \wedge \frac{L(\beta'; z; y) \pi(\beta')}{L(\beta; z; y) \pi(\beta)}$$

set $\beta_{t+1} = \beta'$ else $\beta_{t+1} = \beta$.

end for.

max(r)

(A) S memc $z_t = (z_{1,t} \dots z_{p,t})$ etc. tot. T

$$p_j = \Pr(\text{effect due to } x_j | y) = E(z_j | y)$$

→ estimate post pros

$$\hat{p}_j = \frac{1}{T} \sum_{t=1}^T z_{j,t}$$

accept $z_j = 1$ if $\hat{p}_j$ is large enough:

(B). mode of z (most frequently occurring)
→ is estimated using
Set of z -values in memc output

(C) Estimate $E(p_{ij} | y)$ using posterior averaged
row models

$$\text{let } E_{ij} = E(p_{ij} | y)$$

$$\hat{E}_{ij} = \frac{1}{T} \sum_{t=1}^T p_{ij}(z_t, \beta_t)$$

$$p_{ij}(z_t, \beta_t) = \frac{e^{x_i(\beta_t)z_t}}{e^{x_i(\beta_t)z_t} + e^{x_j(\beta_t)z_t}}$$

{post mean since loss is MSE}

Mock 1 Q2.

2(a) $DI \Leftrightarrow \pi(\theta) f(y|\theta)$ cannot be calculated in reasonable time.

(i) Can't do MH-MCMC in standard form as ratios $\frac{\pi(\theta') f(y|\theta')}{\pi(\theta) f(y|\theta)}$ must be evaluated rapidly.

(ii) Let $X_t = \theta$. Proposal $q(\theta'|\theta)$ given (satisfies $q(\theta'|\theta) > 0 \Leftrightarrow q(\theta|\theta') > 0$).

① Sample $\theta' \sim q(\theta'|\theta)$, $y' \sim f(y'|\theta')$

② Accept if $[d(S(y'), S(y)) < \rho \text{ and}$

$$U(0,1) \leq \min\left(1, \frac{\pi(\theta')}{\pi(\theta)}\right)$$

$q(\theta|\theta')/q(\theta'|\theta)$ ratio is missing here.

Set $X_t = \theta'$ else $X_t = \theta$.

(b) (i) The prior has a norm. const

$$\pi(\psi, \phi|\theta) = \frac{\tilde{\pi}(\psi, \phi|\theta)}{c(\theta)}$$

$$c(\theta) = \frac{\prod_{i=1}^n \tilde{\pi}_i}{\prod_{i=1}^m \tilde{\pi}_i} \int \int \tilde{\pi}(\psi, \phi|\theta) d\psi d^m \phi$$

∴ this integral is intractable so

$$\pi(\theta, \phi, \psi | y) \propto L(y; \phi, \psi) \pi(\phi, \psi | \theta) \pi(\theta)$$

and $\pi(\phi, \psi | \theta)$ is untractable for θ .

$$\pi(\theta | y) = \int \int L(y; \phi, \psi) \pi(\phi, \psi | \theta) d\phi d\psi, \quad \pi(\theta)$$

$$L(y; \phi, \psi) = \prod_{i=1}^m \prod_{j=1}^n (\phi_i \psi_j)^{y_{ij}} e^{-\phi_i \psi_j} \quad \xrightarrow{L(y; \theta)}$$

$$\pi(\phi, \psi | \theta) = \frac{\prod_{i=1}^m \pi(\phi_i, \psi | \theta)}{c(\theta)}, \quad \pi \text{ given in Q.}$$

2(b)ii The mean is suff. stat for Poisson mean. Thus if say $\psi = \text{const}$ over columns then $\sum_{j=1}^n y_{ij}$ is suff stat for ϕ_i

2(b)iii Consider $\pi(\theta | y) \propto L(y; \theta) \pi(\theta)$
 $\uparrow$ given above.

We can simulate $y \sim L(y; \theta)$ by simulating

~~$\theta \sim \pi(\theta)$~~ $\phi, \psi \sim \pi(\phi, \psi | \theta)$ (given MCMC)

then $y \sim L(y; \phi, \psi)$ (Poisson)

$\Rightarrow$ algorithm is same as 2(a)ii except at step ① $\theta' \sim q(\theta | \theta)$, $(\phi', \psi') \sim \pi(\phi', \psi' | \theta')$
 for $j=1, \dots, n$ $y_{ij} \sim \text{Poi}(\phi'_i \psi'_j)$ (i.e. $i=1, \dots, m$)

(only approx at ϕ', ψ' won't have exact $\pi(\phi, \psi | \theta)$ due as MCMC.

$$3(a)(i) \quad p(x_1, \dots, x_n) = p(x_{\sigma_1}, \dots, x_{\sigma_n})$$

$\sigma_1, \dots, \sigma_n$ any permutation of $1, \dots, n$.

$$(ii) \quad p(x_1, \dots, x_n) = \int_0^1 \prod_{i=1}^n q^{x_i} (1-q)^{1-x_i} f(dq)$$

$$(iii) \quad p(x_1, \dots, x_n | n_0) = \frac{p(x_1, \dots, x_n)}{p(n_0)}$$

$$p(n_0) = \sum_{x_1, \dots, x_n: \sum x_i = n_0} p(x_1, \dots, x_n)$$

$$= \binom{n}{n_0} p(\underbrace{1, \dots, 1}_{n_0}, \underbrace{0, \dots, 0}_{n-n_0})$$

now $x_1, \dots, x_n$ satisfy $\sum x_i = n_0$ so

$x_1, \dots, x_n$ is also perm. of n_0 1's, $n-n_0$ 0's

so

$$\begin{aligned} p(x_1, \dots, x_n | n_0) &= \frac{p(\overbrace{1, \dots, 1}^{n_0}, \underbrace{0, \dots, 0}_{n-n_0})}{\binom{n}{n_0} p(\underbrace{1, \dots, 1}_{n_0}, \underbrace{0, \dots, 0}_{n-n_0})} \\ &= \binom{n}{n_0}^{-1} \end{aligned}$$

(b) $x_1, \dots, x_n$ is transformed #2 of

them are $0 \leftrightarrow 1$

$$p(x_1, \dots, x_n) = \frac{1}{2} a^{\sum x_i} (1-a)^{n-\sum x_i}$$

$$= \exp(-\log(2) + \sum x_i \log(a) + (n-\sum x_i) \log(1-a))$$

$$(ii). p(1, 0 \dots 0) = \frac{1}{2} a (1-a)^{n-1}$$

$$p(0, 1, 0 \dots 0) = \frac{1}{2} a^2 (1-a)^{n-2}$$

$$\frac{1}{2} a^2 (1-a)^{n-2} \geq \frac{1}{2} a (1-a)^{n-1}$$

$\Rightarrow a \in \{0, 1\}, \frac{1}{2}$ is necessary.

$$a=0 \quad p(1 \dots 1) = p(0 \dots 0) = \frac{1}{2}$$

$p(x_1 \dots x_n) \leq 0$ otherwise
 $\Rightarrow$ exchangeable.

$a = \frac{1}{2}$ for IID $\Rightarrow$ exchangeable.

$a = 1$ alternating $1010 \dots$ prob $= \frac{1}{2}$
 $0101 \dots$

perms of these prob $> 0 \Rightarrow$ not exch.

$X_0, X_1, X_2 \dots$ exchangeable iff $a \in \{0, \frac{1}{2}\}$.

$$3 (c) ii) Z = (Z_1, \dots, Z_n) \quad Z \sim N(0, \Sigma^{(n)})$$

$$\Sigma^{(n)} = \begin{pmatrix} 1 & \rho \\ & \ddots & \ddots \\ & & 1 & \rho \\ & & & \ddots & \ddots \\ & & & & 1 \end{pmatrix}$$

$$i) \quad \theta \sim N(0, \rho) \quad , \quad Z_i | \theta \sim N(\theta, 1-\rho) \quad \text{i.i.d. } i=1 \dots n$$

$$\Rightarrow E(Z_i) = 0 \quad , \quad \text{var}(Z_i) = 1 \quad \text{cov}(Z_i, Z_j) = \rho$$

$$\Rightarrow Z \sim N(0, \Sigma^{(n)}) \quad (\text{lin comb. of normals is normal})$$

3 (c) ii $p_n(z_1, \dots, z_n)$ exists $\forall n = 1, 2, \dots$

Take $\begin{pmatrix} 1 & \dots & p \\ p & & 1 \end{pmatrix}$ need to show all evals

are +ve. $(1, \dots, 1)$ is eigenvector eval $1 + (n-1)p$

$(1, 0, \dots, 0, -1, 0, \dots, 0)$ is $\perp$ $n-1$ eigenvectors

eval $1-p$ so we see p_n exists iff

$$1 + (n-1)p > 0 \quad \forall n$$

$$\Leftrightarrow p > \frac{-1}{n-1} \quad \forall n$$

$$\Leftrightarrow p \geq 0$$

$$\text{and } 1-p > 0$$

$$p < 1$$

ie $0 \leq p < 1$ is N+S.

3 (c) iii

$$\int_{-\infty}^{\infty} p_{n+1}(z_1, \dots, z_{n+1}) dz_{n+1}$$

$$= \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} \prod_{i=1}^{n+1} N(z_i; \theta, 1-p) N(\theta; 0, p) d\theta \right] dz_{n+1}$$

$$= \int_{-\infty}^{\infty} \prod_{i=1}^n N(z_i; \theta, 1-p) N(\theta; 0, p) d\theta$$

exchanging integrals

$$= p_n(z_1, \dots, z_n)$$

3(c) ~~3(v)~~

$$\begin{aligned} p(x_1, \dots, x_n) &= \int p(x_1, \dots, x_n | z_1, \dots, z_n) p(z_1, \dots, z_n) dz_1, \dots, dz_n \\ &= \int \prod_i [\phi(z_i)^{x_i} (1 - \phi(z_i))^{1-x_i} f(z_i | \theta)] dF(\theta) dz_1, \dots, dz_n \\ &\quad \times dz_1, \dots, dz_n \end{aligned}$$

$$= \int \prod_{i=1}^n g(x_i | \theta) dF(\theta)$$

$$g(x_i | \theta) = \int \phi(z_i)^{x_i} (1 - \phi(z_i))^{1-x_i} f(z_i | \theta) dz_i$$

$$f(z_i | \theta) = N(z_i; \theta, 1 - \rho)$$

$$dF(\theta) = N(\theta; 0, \rho) d\theta.$$

} So $x_1, \dots, x_n, \dots$ exchangeable
by de F.